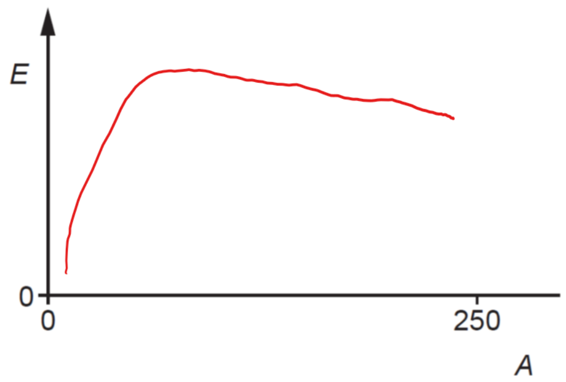
Qns	Answer	Marks
5(a)(i)	energy required to separate the nucleons (in the nucleus) to infinity	**M1** **M1**
5(a)(ii)	curve starting close to the origin and forming a single peak peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak  **Fig. 5.1**	**B1**
5(b)(i)	Fusion	**B1**
5(b)(ii)	both particles have low *A* values or both particles are at left-hand end of graph He-3 has higher binding energy (per nucleon) than the total binding energy of both H-2	**B1** **B1**
5(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)]$ u $= 0.00351$ u (Correct substitution and correct answer) $E = \Delta mc^2$ $= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 5.24 \times 10^{-13}$ J (Correct substitution and correct answer) 1.0 mol of deuterium forms 0.500 mol of helium-3 total energy $= 0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ (Correct substitution) $= 1.58 \times 10^{11}$ J	 **C1** **C1** **C1** **C1** **A1**